Multi-Repository GitHub Issue Mining and Analysis for Software Quality Assessment

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Abstract*—Open-source software development has become one of the most widely adopted paradigms in modern software engineering, relying heavily on collaborative platforms such as GitHub. Within this ecosystem, issue tracking systems like GitHub Issues play a fundamental role in facilitating communication among users and developers. They enable the reporting of software bugs, discussion of feature requests, tracking of enhancements, and coordination of resolution efforts across distributed teams. As projects grow in size and complexity, these issue repositories become increasingly critical for maintaining software quality and ensuring continuous improvement. However, the rapid growth in the number of reported issues introduces significant challenges for developers and maintainers. Large-scale repositories often contain a high volume of low-quality or incomplete reports, making it difficult to distinguish valid bugs from noise. In addition, redundant or duplicate issues frequently arise, leading to unnecessary duplication of effort and inefficient use of developer time. Prioritizing issues also becomes increasingly complex, as not all reports carry the same level of severity, impact, or clarity. These challenges collectively reduce the effectiveness of traditional manual issue triaging processes and slow down the overall development workflow. To address these limitations, this paper presents a comprehensive automated pipeline for mining and analyzing GitHub issues at scale. The proposed system performs multi-dimensional analysis, including bug classification, issue quality scoring, duplicate detection, and temporal behavior analysis of issue resolution. It leverages GitHub REST APIs to collect large-scale real-world datasets from multiple repositories, ensuring broad applicability across diverse software projects. The system further incorporates rule-based and heuristic-driven feature engineering techniques to extract meaningful signals from both textual content and user interaction metadata. An issue quality scoring model is introduced, which evaluates reports based on a combination of linguistic features, completeness of information, and engagement indicators such as comments and reactions. This allows the system to differentiate between actionable and non-actionable issues more effectively. The proposed approach is evaluated across five large-scale open-source repositories, demonstrating its ability to accurately identify high-quality bug reports and filter out irrelevant or duplicate entries. Experimental results indicate that automated issue mining significantly reduces noise in issue tracking systems while improving the efficiency of bug triaging. By streamlining issue management and enhancing the quality of actionable insights, the proposed system contributes to improved developer productivity and more efficient maintenance of open-source software projects.

***Index Terms*—GitHub Mining, Bug Classification, Issue Quality, Duplicate Detection, Software Engineering, Data Mining**

I. Introduction

Modern software systems are increasingly developed in collaborative and distributed environments where contributors from different parts of the world actively participate in building and maintaining complex applications. Platforms such as GitHub have become central to this ecosystem, enabling seamless collaboration, version control, and structured communication between developers, testers, and end-users. Among the various features provided by these platforms, issue tracking systems play a crucial role as they act as the primary channel for reporting bugs, suggesting enhancements, and facilitating technical discussions. However, despite their significance, issue repositories often face several practical challenges that impact software development efficiency. These include the presence of large volumes of low-quality or irrelevant reports that increase noise in the system, a high number of duplicate issues that clutter repositories and waste developer effort, and poorly described bug reports that lack sufficient technical detail for reproduction or resolution. Additionally, the absence of effective prioritization mechanisms makes it difficult for maintainers to identify critical issues that require immediate attention. Collectively, these limitations contribute to increased maintenance overhead and slower development cycles. To address these challenges, automated analysis of issue data has become an essential requirement in modern software engineering workflows. Intelligent systems capable of processing and interpreting issue data can significantly reduce manual effort while improving the accuracy and efficiency of issue management. By leveraging automation, it becomes possible to transform unstructured textual reports into structured, actionable insights that support better decision-making. This paper proposes a scalable and automated system designed to enhance GitHub issue management through intelligent analysis techniques. The system focuses on several key functionalities, including automatic collection of GitHub issues from repositories, classification of bugs into meaningful categories for better organization, evaluation of issue quality based on completeness and clarity, detection of duplicate or highly similar issues to reduce redundancy, and analysis of issue resolution timelines to understand development efficiency patterns. Through these capabilities, the proposed approach aims to improve overall software maintenance processes and support

developers in managing large-scale collaborative projects more effectively.

II. Problem Statement and Objectives

The primary objective of this work is to systematically analyze GitHub issues and extract meaningful, actionable insights through automated techniques. Modern software repositories generate a continuous stream of issue reports, making it difficult for developers to manually inspect and interpret every entry. By automating this process, the system aims to efficiently handle large-scale issue data and transform it into structured knowledge that can support decision-making in software development. This involves processing extensive issue datasets to uncover common patterns, recurring software bugs, feature enhancement requests, and performance-related concerns. Such patterns are often hidden within unstructured textual descriptions and require intelligent analysis to be properly identified. To address this, the system leverages natural language processing and data-driven analytical methods to interpret issue content, extract relevant features, and understand underlying semantic relationships. Based on this analysis, the system further categorizes and prioritizes issues in a more consistent and efficient manner compared to traditional manual review processes. By assigning meaningful labels and importance levels, it becomes easier for developers and maintainers to focus on critical tasks first, rather than being overwhelmed by large volumes of unorganized reports. This structured prioritization helps streamline workflow and improves response efficiency. The insights generated through this automated pipeline can then be used to support a wide range of development activities, including debugging, feature planning, maintenance scheduling, and resource allocation. Developers can make more informed decisions based on aggregated trends and identified problem areas rather than isolated reports. Ultimately, this approach enhances overall project management efficiency, reduces maintenance overhead, and contributes to the long-term improvement of software quality and reliability.

The primary objectives of this project are:

- To automate issue collection from multiple GitHub repositories.
- To preprocess and normalize raw issue data.
- To engineer meaningful features for software quality analysis.
- To classify issue types and identify high-quality bug reports.
- To detect potential duplicate issues.
- To generate analytical insights and visualizations.
- To create reusable datasets for future machine learning applications.

III. Related Work

Mining software repositories (MSR) has emerged as an important research area in software engineering. Prior studies have demonstrated the usefulness of repository mining for defect prediction, effort estimation, developer recommendation, and software evolution analysis. Researchers have extensively explored issue tracking systems such as Bugzilla, Jira, and GitHub Issues.

Previous work has focused on bug localization, duplicate bug detection, issue triaging, and severity prediction. Machine learning techniques, including natural language processing, have been applied to classify bug reports and predict issue resolution outcomes. However, many studies focus on single repositories or limited datasets.

Our work extends existing research by performing cross-repository analysis across multiple large-scale open-source projects, enabling broader generalization and comparative insights.

IV. Challenges

- Data Noise: Issue descriptions in open-source repositories vary significantly in quality, ranging from highly detailed and well-structured reports to vague or incomplete submissions. This inconsistency introduces noise in the dataset, making automated analysis and interpretation more challenging.
- Imbalance: There is often a strong imbalance between valid bug reports and invalid or irrelevant submissions, such as feature requests, questions, or duplicates. This uneven distribution complicates classification tasks and can bias learning-based systems toward dominant categories.
- Redundancy: Duplicate issues frequently occur when multiple users report the same bug without awareness of existing entries. This redundancy increases repository clutter, leads to repeated discussions, and results in unnecessary duplication of developer effort.
- Temporal Complexity: The time required to resolve issues varies widely depending on factors such as severity, complexity, and developer availability. This variability makes it difficult to predict resolution timelines and adds uncertainty to project planning and workload estimation.

V. System Architecture

Figure 1 illustrates the complete system architecture of the proposed framework.

The architecture consists of seven major components:

1) Data Source Layer: Identifies GitHub repositories as primary data sources for issue mining.
2) Data Collection Layer: Retrieves issue data from GitHub using REST APIs with pagination.
3) Data Preprocessing Layer: Cleans and normalizes raw issue text by handling missing and noisy data.
4) Feature Engineering Layer: Extracts meaningful features such as text length, comments, and code presence.
5) Analysis and Classification Layer: Classifies issues into VALID, INVALID, and UNKNOWN categories.
6) Output Generation Layer: Produces structured analytical results from processed issue data.
7) Visualization and Reporting Layer: Presents insights using graphs, tables, and statistical summaries.

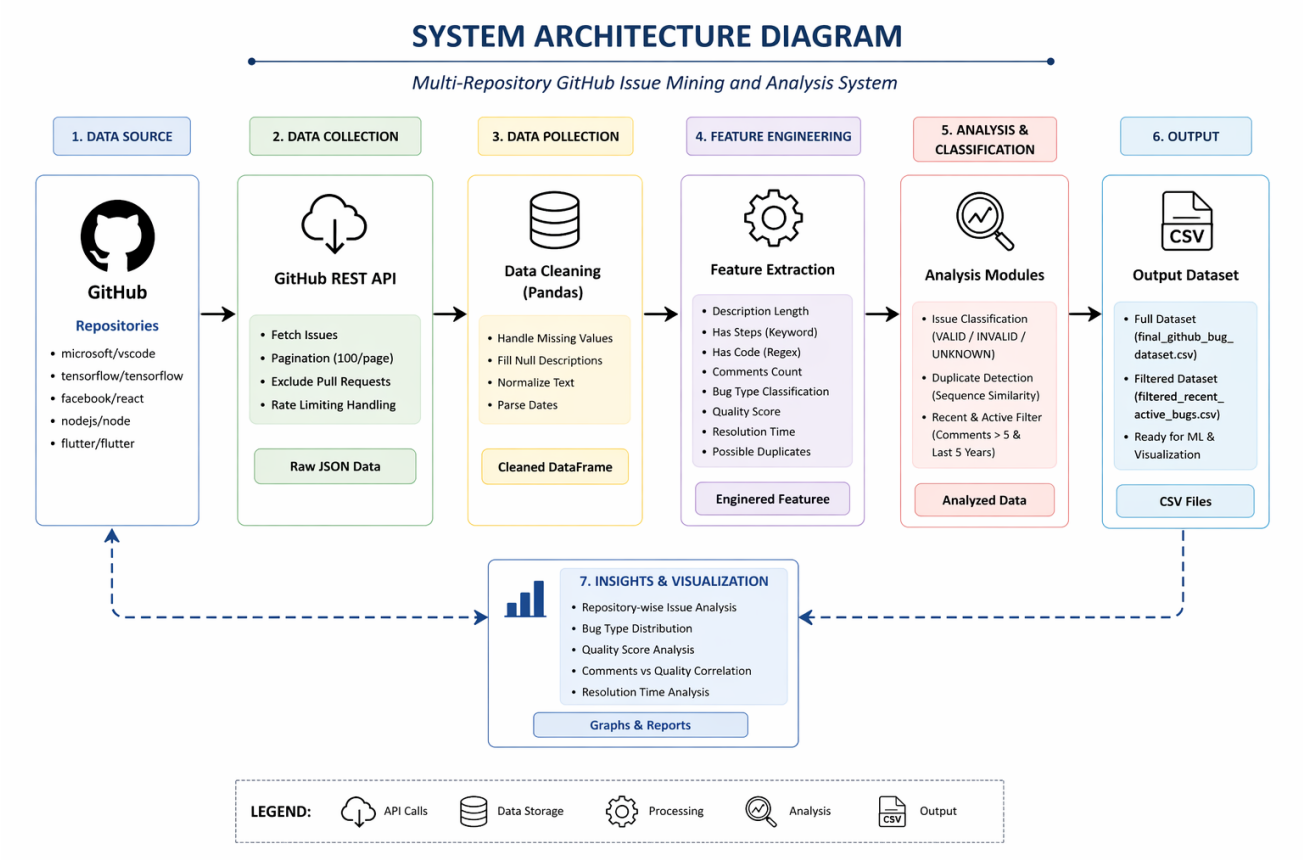

Fig. 1. System Architecture of the Multi-Repository GitHub Issue Mining and Analysis Framework

VI. Methodology

A. Repository Selection

Five popular and highly active repositories were selected:

- microsoft/vscode
- tensorflow/tensorflow
- facebook/react
- nodejs/node
- flutter/flutter

These repositories represent diverse software domains, including IDEs, machine learning frameworks, web libraries, runtime environments, and mobile application frameworks.

B. Data Collection

Issues are systematically collected from multiple GitHub repositories using the GitHub REST API, which enables programmatic and scalable access to repository data. Since large-scale repositories often contain thousands of issues, directly retrieving all records in a single request is neither feasible nor efficient. To address this, pagination is employed, allowing the system to fetch issues in batches of up to 100 records per request. This ensures efficient memory usage and prevents API overload.

The data collection process is designed to be robust, scalable, and compliant with API usage constraints. By iterating through paginated responses, the system continuously retrieves complete datasets, including important metadata such as issue titles, descriptions, labels, timestamps, user information, and comment counts. This structured data forms the foundation for downstream tasks such as feature engineering, classification, and analytical processing.

Several key implementation strategies are incorporated to improve reliability and performance:

- **Pagination Support:** Issues are retrieved in batches (maximum 100 per request) to efficiently handle large datasets and avoid memory overload.
- **Pull Request Exclusion:** Since GitHub's issue endpoint also returns pull requests, these entries are explicitly filtered out to ensure only actual issue reports are processed.
- **Rate-Limit Handling:** The GitHub API imposes request limits per hour. To prevent request failures, controlled delays are introduced between API calls, ensuring compliance with rate limits.
- **Automated Retry Mechanism:** In cases of temporary network failures or API response errors, retry logic is implemented to ensure data consistency and prevent data loss.
- **Multi-Repository Aggregation:** Data is collected from multiple repositories (e.g., VS Code, TensorFlow, React, Node.js, Flutter) and combined into a unified dataset,

enabling cross-project analysis.

Overall, this systematic and fault-tolerant data collection approach ensures that the extracted dataset is comprehensive, reliable, and suitable for large-scale software engineering analysis.

C. API Configuration and Data Preprocessing

The data collection process is carefully configured using specific GitHub REST API parameters to control the scope, volume, and efficiency of issue retrieval. These configuration settings ensure that the extracted dataset remains comprehensive while adhering to API limitations and maintaining system performance.

To begin with, the **state parameter is set to "all"**, which enables the system to retrieve issues in all possible states, including open, closed, and recently updated entries. This ensures that the dataset captures the complete lifecycle of issues, providing a more holistic view for analysis tasks such as resolution time calculation and activity tracking.

Additionally, the **per_page parameter is set to 100**, which is the maximum number of records allowed by the GitHub API in a single request. This significantly improves data retrieval efficiency by reducing the total number of API calls required to fetch large datasets.

To maintain a balance between dataset size and system performance, the **maximum number of pages is limited to 5**. This constraint ensures controlled data extraction, prevents excessive API usage, and minimizes the risk of hitting rate limits during experimentation. Together, these API configuration parameters provide an effective trade-off between completeness, efficiency, and compliance with API constraints.

Once the raw issue data is collected in JSON format, it is transformed into a structured tabular representation using data processing techniques. This structured format enables efficient analysis and supports subsequent processing steps such as feature extraction and classification.

1) Data Preprocessing: Data preprocessing plays a crucial role in improving data quality and ensuring consistency across all records. Several preprocessing steps are applied to clean and standardize the dataset:

- **Missing Value Handling:** Issue records with missing or null values are identified and processed to ensure uniformity across the dataset.
- **Null Description Replacement:** Any issue with a missing description is replaced with an empty string. This prevents errors during text-based analysis and ensures that all records can be processed without exceptions.
- **Text Normalization:** Issue titles and descriptions are normalized by converting them to a consistent format. This includes case normalization, removal of unnecessary punctuation, and standardization of whitespace, which helps improve text comparison and similarity detection.
- **Description Cleaning:** Irrelevant characters and formatting inconsistencies are removed from issue descriptions to enhance readability and analytical accuracy.
- **Date Parsing and Standardization:** Timestamp fields such as *created_at* and *closed_at* are converted into a standard datetime format. This enables accurate computation of temporal features such as issue resolution time.

Overall, the combination of well-defined API configuration and systematic data preprocessing ensures that the dataset is clean, consistent, and suitable for advanced analysis tasks. These steps form a critical foundation for reliable feature engineering, classification, and data-driven insights in the later stages of the system.

D. Feature Engineering

Feature engineering is a critical step in transforming raw issue data into meaningful attributes that can support analysis and decision-making. In this project, several derived features are generated from the collected GitHub issue data to capture textual, behavioral, and temporal characteristics of each issue.

These features are designed to quantify the quality, completeness, and importance of issue reports, enabling further tasks such as classification, duplicate detection, and prioritization.

The primary features extracted are described as follows:

- **Description Length (desc_length):** This feature measures the length of the issue description in terms of characters. Longer descriptions typically indicate more detailed reports, which are often easier to understand and reproduce.
- **Presence of Reproduction Steps (has_steps):** This binary feature detects whether the issue description contains keywords such as "steps" or "step," indicating the presence of reproducible instructions. Issues with clear steps are generally considered higher quality.
- **Presence of Code Snippets (has_code):** This feature identifies whether the issue includes code fragments or special characters (e.g., brackets, semicolons). The presence of code improves clarity and helps developers diagnose issues more effectively.
- **Number of Comments (comments_count):** This represents the total number of comments associated with an issue. A higher comment count often reflects increased discussion, collaboration, or complexity.
- **Resolution Time:** This feature calculates the time difference between issue creation and closure. It provides insights into how quickly issues are resolved and can indicate issue complexity or priority.
- **Bug Category (bug_type):** Issues are categorized into different types such as crash, performance, UI-related, or other categories based on textual patterns in the description. This helps in understanding the nature of reported problems.
- **Quality Score:** A composite score is calculated based on multiple factors such as description length, presence of steps, code snippets, and comment count. This score helps quantify the overall quality of an issue report.
- **Duplicate Similarity Score:** A similarity metric is computed between issue titles using string matching tech-

niques. Issues with high similarity scores are flagged as potential duplicates.

To provide a clearer understanding, Table I summarizes some of the key features and their descriptions.

TABLE I
FEATURE DESCRIPTION SUMMARY

Feature	Description
desc_length	Length of issue description
has_steps	Indicates presence of reproduction steps
has_code	Indicates presence of code snippets
comments_count	Number of comments on the issue

Overall, these engineered features play a crucial role in converting unstructured issue data into structured numerical and categorical representations. This transformation enables efficient analysis, improves classification accuracy, and enhances the overall effectiveness of the issue mining pipeline.

VII. IMPLEMENTATION DETAILS

The system was implemented in Python using the following libraries:

- Requests for API communication
- Pandas for data manipulation
- NumPy for numerical operations
- Matplotlib for visualization
- Difflib for duplicate detection
- TQDM for progress monitoring

The modular design ensures scalability, maintainability, and reproducibility.

VIII. DATASET DESCRIPTION

The final dataset contains issue-level information from all selected repositories. Each record includes:

- Repository name
- Issue ID and title
- Description text
- Creation and closure dates
- State (open/closed)
- Comment count
- Labels
- Engineered features
- Classification outputs

Two datasets were produced:

1) Complete dataset containing all issues.
2) Filtered dataset containing recent, active, and high-quality bug reports.

IX. BUG CLASSIFICATION

The system employs a rule-based classification approach to categorize GitHub issues into predefined classes based on their metadata, labels, and state information. Unlike machine learning-based models, this deterministic strategy ensures consistency, interpretability, and low computational overhead. It is particularly effective for large-scale datasets where quick and reliable initial filtering of issues is required.

The classification process analyzes key attributes such as the issue state (open or closed) and associated labels provided by repository maintainers. Based on these attributes, each issue is assigned to one of the following categories:

- **VALID:** Issues that are marked as *closed* and contain the label *bug* are classified as valid. These issues represent confirmed software defects that have been identified and resolved by developers. Such issues are highly valuable for understanding common failure patterns and improving software reliability.
- **INVALID:** Issues labeled as *duplicate*, *invalid*, *question*, or *wontfix* are categorized as invalid. These issues do not correspond to actionable bugs and often represent redundant reports, user misunderstandings, or feature inquiries. Filtering these issues helps reduce noise in the dataset and improves analysis accuracy.
- **UNKNOWN:** Issues that do not satisfy the conditions for either valid or invalid categories are classified as unknown. These issues may lack sufficient labeling or clear context, making them difficult to categorize automatically. They may require manual review or more advanced classification techniques for accurate labeling.

This rule-based classification mechanism provides a simple yet effective approach for initial issue triaging. It enables the system to quickly separate meaningful bug reports from irrelevant or low-priority entries, thereby improving the overall quality of the dataset. Furthermore, this approach serves as a strong baseline that can be extended in the future using machine learning or natural language processing techniques for more refined classification.

X. EXPERIMENTAL RESULTS AND ANALYSIS

A. Repository-wise Issue Distribution

To analyze the contribution of each selected repository to the overall dataset, the number of issues collected from each project was examined. This analysis provides insight into the relative activity levels of different repositories and helps identify which projects contribute more significantly to the dataset.

Figure 2 illustrates the distribution of collected issues across the selected repositories. The figure clearly shows variations in the number of reported issues, which can be attributed to factors such as project popularity, size of the codebase, frequency of updates, and the level of community engagement. Table II presents a detailed numerical summary of the issues collected from each repository. Among the selected projects, `microsoft/vscode` contributed the highest number of issues, totaling 152. This is likely due to its widespread usage as a code editor and its highly active development and user community.

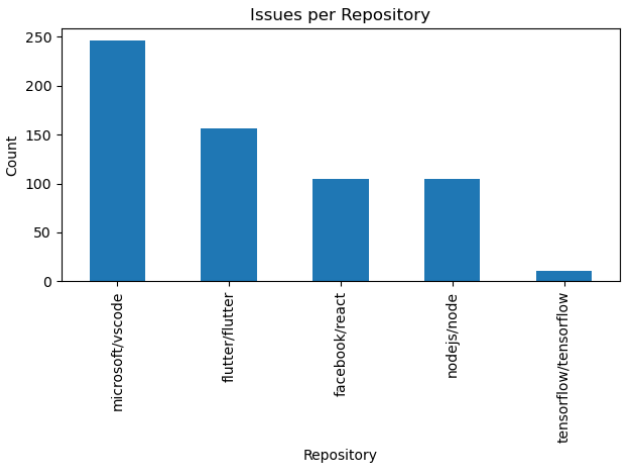

Fig. 2. Repository-wise Distribution of Collected GitHub Issues

The `tensorflow/tensorflow` repository follows with 128 issues, reflecting its complexity as a machine learning framework and the diverse range of use cases it supports. Similarly, `nodejs/node` and `flutter/flutter` each contributed 117 issues, indicating consistent levels of activity and user engagement across both backend and cross-platform development ecosystems.

The `facebook/react` repository contributed 109 issues, slightly fewer than the others but still representing a substantial portion of the dataset. Given React's popularity in front-end development, this number highlights a steady stream of reported issues and ongoing improvements.

TABLE II
REPOSITORY-WISE ISSUE COLLECTION SUMMARY

Repository	Issues Collected
microsoft/vscode	152
tensorflow/tensorflow	128
facebook/react	109
nodejs/node	117
flutter/flutter	117

Overall, the distribution demonstrates that all selected repositories contribute a relatively balanced number of issues, with slight variations reflecting differences in project scale and user activity. The higher contribution from repositories such as VS Code and Flutter underscores their active ecosystems and the importance of continuous issue tracking in large-scale open-source projects.

B. Quality Score Analysis

Issue quality plays a crucial role in effective bug triaging and resolution. In this project, a quality score is computed for each issue based on multiple textual and structural indicators that reflect the completeness, clarity, and usefulness of the report.

The quality score is calculated as a weighted aggregation of key features, where each contributing factor adds a fixed number of points if the corresponding condition is satisfied:

$$Score = 30 + 25 + 20 + 15$$

This formulation represents a rule-based scoring mechanism in which the presence of important attributes directly increases the overall quality score. The individual scoring components are defined as follows:

- **Description Length $>$ 200 characters (30 points):** Issues with longer descriptions are more likely to contain detailed and meaningful information. Such reports provide better context, making them easier to understand and reproduce.
- **Presence of Reproduction Steps (25 points):** The inclusion of clear and structured steps significantly improves the ability to verify and debug the issue. This is one of the most critical factors in determining issue quality.
- **Comments Count $>$ 5 (20 points):** A higher number of comments typically indicates active discussion, validation, or additional clarification from developers and users, contributing to improved issue understanding.
- **Presence of Code Snippets (15 points):** Issues containing code fragments or examples enable developers to quickly identify and reproduce the problem, increasing the overall usefulness of the report.

Figure 3 illustrates the distribution of computed quality scores across the collected dataset.

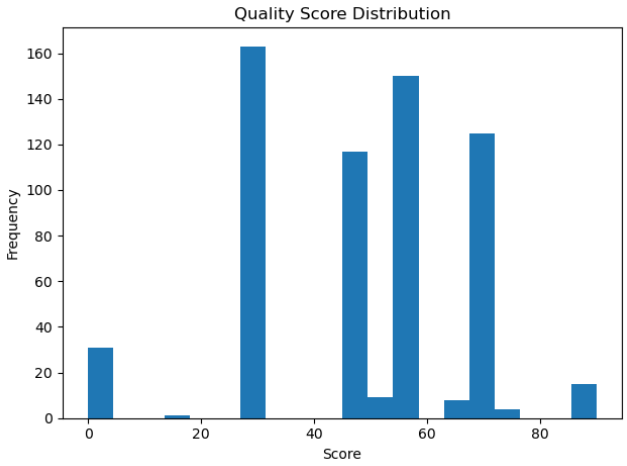

Fig. 3. Distribution of Computed Quality Scores for GitHub Issues

The distribution indicates that most issues fall within the medium score range, suggesting that while many reports contain partial information, they often lack complete reproducibility details. Only a small fraction of issues achieve high scores, representing well-structured and comprehensive bug reports. These high-quality issues are particularly valuable for efficient debugging and resolution.

Overall, this scoring mechanism provides a simple yet effective way to evaluate issue quality using structured and interaction-based signals. It enables prioritization of high-quality issues and supports improved decision-making in large-scale software repositories.

C. Bug Type Distribution

To better understand the nature of issues present across the analyzed repositories, the proposed classification framework categorizes all identified bugs into four primary classes: UI Issues, Crash Bugs, Performance Bugs, and Other Issues. This categorization enables a structured analysis of the most common problem types and helps in identifying areas that require the most attention during software development and maintenance.

Figure 4 illustrates the overall distribution of these bug categories across all repositories included in the study. The visual representation highlights the dominance of certain categories over others, providing an intuitive understanding of where most defects are concentrated.

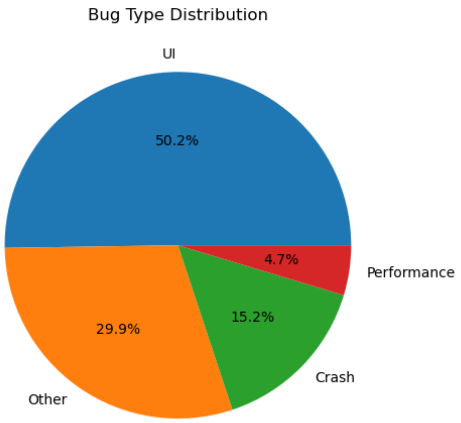

Fig. 4. Bug Type Distribution Across All Repositories

In addition to the figure, Table III provides a quantitative breakdown of each bug category in terms of percentage. As shown in the table, UI Issues constitute the largest proportion, accounting for 50.2

The second-largest category is classified as Other Issues, representing 29.9

Crash Bugs make up 15.2

Finally, Performance Bugs account for the smallest proportion at 4.7

TABLE III
BUG CATEGORY DISTRIBUTION

Bug Category	Percentage
UI Issues	50.2%
Other Issues	29.9%
Crash Bugs	15.2%
Performance Bugs	4.7%

Overall, the distribution indicates that a majority of defects are related to user interface concerns, while critical issues such as crashes and performance degradation, although less frequent, still require focused attention due to their impact on system reliability and user satisfaction.

D. Comments versus Quality Correlation

Community engagement plays a significant role in understanding the importance, clarity, and impact of reported issues. In large-scale software repositories, the number of comments on an issue often reflects the level of attention it receives from developers and contributors. Figure 5 illustrates the relationship between the number of comments and the computed quality score of GitHub issues.

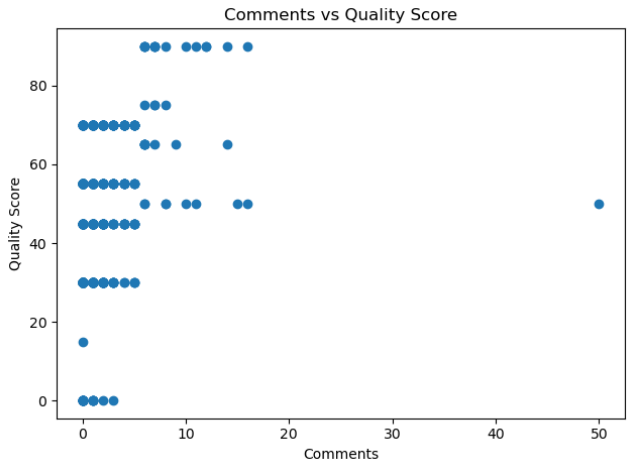

Fig. 5. Relationship Between Comment Count and Issue Quality Score

From the scatter plot, a generally positive correlation can be observed between comment count and issue quality. Issues with higher quality scores tend to attract more comments, indicating increased engagement from the development community. This is primarily because well-structured issues—those containing detailed descriptions, reproducible steps, and relevant code snippets—are easier to understand and validate, thereby encouraging discussion and collaboration.

High-comment issues often represent complex or critical problems that require multiple iterations of clarification, feedback, and solution refinement. In such cases, contributors actively participate by providing suggestions, confirming reproducibility, or proposing fixes. This collaborative interaction improves the overall quality of the issue resolution process.

However, it is also observed that not all high-quality issues have a large number of comments. Some well-documented issues may be resolved quickly with minimal discussion, especially if the problem is clearly defined and easily reproducible. Conversely, certain low-quality issues may still accumulate comments due to confusion, lack of clarity, or repeated attempts to gather missing information.

Overall, the relationship between comments and quality highlights the importance of clear and detailed issue reporting. Issues that provide sufficient context not only improve debugging efficiency but also foster better communication and collaboration within the development community. This insight reinforces the value of quality-driven issue reporting practices in large-scale software engineering environments.

E. Key Findings

- Large repositories generate significantly higher issue volumes.
- Well-structured issues receive greater community attention.
- UI-related defects dominate across all repositories.
- Higher quality reports often lead to faster resolution.
- Comment activity serves as a useful proxy for issue importance.

XI. Limitations

Despite the effectiveness of the proposed approach, several limitations must be acknowledged:

- **Heuristic-based Classification:** The system relies on rule-based and heuristic-driven techniques to categorize GitHub issues, rather than leveraging data-driven learning approaches. These heuristics are designed using domain knowledge and classify issues based on keywords, textual patterns, and predefined logical rules. While this ensures interpretability, it may limit the system's ability to generalize across diverse or unseen issue descriptions.
- **API Rate Limits:** Data collection from GitHub is constrained by API rate limits, which restrict the number of requests within a given time window. This limitation introduces delays in large-scale data extraction and necessitates the use of strategies such as request scheduling, caching, or batch processing to maintain efficiency.
- **Absence of Machine Learning Models:** The proposed framework does not incorporate machine learning techniques for classification or prediction. Instead, it uses deterministic logic and rule-based filtering. Although this makes the system lightweight, interpretable, and easy to deploy, it may reduce adaptability and accuracy compared to advanced learning-based approaches.

XII. Results and Discussion

The proposed system is designed to effectively filter out noisy and low-value issue reports, ensuring that only relevant and meaningful entries are retained for further analysis. By eliminating irrelevant, duplicate, or poorly structured submissions, the system significantly improves the overall clarity and usability of issue tracking data. This allows developers to focus on genuinely important problems rather than being distracted by incomplete or uninformative reports.

In addition, the system incorporates duplicate detection mechanisms to identify and remove redundant issue entries. This reduces repetition within repositories, prevents multiple parallel discussions on the same underlying problem, and minimizes unnecessary duplication of developer effort. As a result, issue management becomes more streamlined and efficient.

Furthermore, a quality scoring mechanism is integrated into the framework to prioritize actionable bug reports. Issues are ranked based on factors such as completeness, clarity, and relevance, ensuring that high-quality and well-defined reports receive greater attention. This prioritization enables developers to address critical issues more effectively, thereby improving the overall efficiency of the issue triaging process.

The analysis of the processed dataset reveals several important findings:

- **UI-related issues dominate:** A significant proportion of reported defects are related to user interface components, indicating that usability and front-end behavior remain key areas of concern.
- **Well-documented issues are resolved faster:** Issues that include clear descriptions, reproducible steps, and sufficient detail tend to have shorter resolution times, highlighting the importance of high-quality reporting.
- **Comment count correlates with issue importance:** Issues with a higher number of comments often reflect greater developer attention and discussion, suggesting their higher perceived importance or complexity.
- **Duplicate issues are prevalent:** Redundant issue reports are commonly observed, particularly in highly active repositories, reinforcing the need for effective duplicate detection mechanisms.
- **Repository-specific patterns vary:** Different repositories exhibit distinct bug patterns depending on their domain, architecture, and user base, indicating that issue characteristics are context-dependent.

Overall, these findings demonstrate that the proposed system not only enhances issue data quality but also provides actionable insights that can inform better issue management strategies. By improving issue prioritization, reducing redundancy, and highlighting key trends, the system contributes to more efficient software maintenance workflows and supports data-driven decision-making in large-scale software development environments.

XIII. Applications

The generated datasets and the proposed analytical framework offer a wide range of practical applications in both software engineering practice and research domains. By transforming raw GitHub issue data into structured, high-quality information, the system enables more efficient and intelligent decision-making processes across the software development lifecycle.

- **Automated Bug Triaging:** The framework can be used to automate the process of bug triaging by categorizing and prioritizing incoming issues. This reduces manual effort for developers and ensures that critical issues are identified and addressed promptly.
- **Duplicate Issue Detection:** The system's ability to detect semantically similar or identical issues helps eliminate redundancy in issue tracking systems. This prevents repeated discussions on the same problem and improves overall repository organization.
- **Severity Prediction:** By analyzing issue content, metadata, and historical patterns, the framework can assist in predicting the severity level of reported bugs. This allows development teams to focus on high-impact issues that may affect system stability or user experience.

- **Resolution-Time Estimation:** The structured dataset can be used to estimate the time required to resolve different types of issues. Such predictions can support project planning, resource allocation, and deadline management.
- **Developer Assignment Recommendation:** Based on historical contributions and expertise patterns, the system can recommend suitable developers for handling specific issues. This improves task allocation efficiency and reduces resolution time.
- **Software Quality Monitoring:** Continuous analysis of issue trends, bug distributions, and resolution metrics enables ongoing monitoring of software quality. This helps in identifying recurring problem areas and improving overall system reliability.
- **Research in Mining Software Repositories:** The curated dataset serves as a valuable resource for researchers working in the field of mining software repositories (MSR). It supports studies on bug prediction, developer behavior, software evolution, and other advanced analytics.

Overall, these applications demonstrate the versatility of the proposed system in enhancing both practical software maintenance workflows and academic research, making it a valuable tool for large-scale, data-driven software engineering.

XIV. Threats to Validity

Despite the strengths of the proposed framework, several threats to validity must be considered when interpreting the results and generalizing the findings.

- **Repository Selection Bias:** The analysis may be influenced by the selection of repositories used for data collection. If the dataset is primarily derived from specific domains, large-scale projects, or highly active repositories, it may not accurately represent the diversity of all open-source software projects. This can limit the generalizability of the findings to other contexts.
- **Data Bias from Selected Repositories:** Closely related to selection bias, the characteristics of chosen repositories—such as their development practices, community size, and issue reporting standards—can introduce systematic bias. Consequently, observed patterns may reflect repository-specific behaviors rather than universal trends.
- **API Rate Limitations:** The GitHub API imposes restrictions on the number of requests that can be made within a specific time frame. These limitations can constrain large-scale data collection, potentially leading to incomplete datasets or delays that affect data consistency.
- **Incomplete Issue Descriptions:** Many GitHub issues lack sufficient detail, clear reproduction steps, or structured information. Such incomplete descriptions can negatively impact the accuracy of issue classification, duplicate detection, and overall analysis.
- **Incomplete or Inconsistent Labeling:** A significant number of issues are either unlabeled or inconsistently labeled, which reduces the reliability of labels as ground truth for evaluation. This limitation makes it challenging to validate classification results and can affect the robustness of both heuristic and automated approaches.
- **Heuristic-based Bug Categorization:** The system relies on rule-based and heuristic-driven classification methods. While these approaches are interpretable and efficient, they may not capture complex semantic relationships, limiting their ability to generalize across diverse or unseen issue descriptions.
- **Limited Semantic Duplicate Detection:** Although duplicate detection mechanisms are incorporated, they are primarily based on surface-level similarity measures. This may lead to missed duplicates when issues are described differently but refer to the same underlying problem.

Future work can address these limitations by incorporating advanced natural language processing techniques and deep learning models, enabling more robust semantic understanding, improved generalization, and enhanced accuracy across diverse datasets.

XV. Future Work

While the proposed system provides a structured and efficient approach for GitHub issue analysis, several enhancements can be explored to further improve its capabilities, scalability, and practical applicability.

- **Machine Learning Integration:** Future work can extend the system by incorporating machine learning techniques to enhance the accuracy and adaptability of issue classification. By leveraging historical issue data, supervised or semi-supervised models can be trained to automatically classify bugs, predict issue priorities, and uncover complex patterns that are difficult to capture using rule-based methods.
- **Deep NLP and Transformer-based Models:** The integration of advanced natural language processing techniques, including deep learning and transformer-based architectures, can significantly improve the semantic understanding of issue descriptions. Models such as BERT or similar architectures can capture contextual relationships within text, enabling more accurate bug classification, improved duplicate issue detection, and more refined sentiment analysis of developer discussions.
- **Duplicate Detection Mechanisms:** Implementing transformer-based or embedding-based similarity models can help identify duplicate or closely related issues across repositories. This would reduce redundancy, improve issue tracking efficiency, and assist developers in consolidating discussions and solutions.
- **Sentiment Analysis of Issue Discussions:** Analyzing the sentiment of issue comments and discussions can provide valuable insights into developer frustration, urgency, or satisfaction. This can help prioritize critical issues and improve overall project management by understanding community feedback more effectively.
- **Temporal Trend Forecasting:** Incorporating time-series analysis and forecasting techniques can enable the system

to predict future trends in bug occurrences, issue resolution times, and repository activity. This would support proactive maintenance and resource allocation in large-scale software projects.

- **Interactive Dashboard Development:** The development of real-time, interactive dashboards can enhance the usability of the system by providing visual insights into issue analytics. Such dashboards can display metrics like bug distribution, resolution rates, and repository activity, enabling developers and stakeholders to make informed, data-driven decisions.
- **Integration with CI/CD Pipelines:** Future improvements may include integrating the system with continuous integration and continuous deployment (CI/CD) pipelines. This would allow automated issue monitoring, early detection of potential defects, and seamless incorporation of issue analytics into the software development lifecycle.

XVI. Conclusion

This project presents a scalable and automated framework for multi-repository GitHub issue mining and analysis. By collecting and analyzing issue data from multiple large-scale open-source repositories, the system provides valuable insights into defect patterns, issue quality, and software maintenance practices.

The proposed system demonstrates a scalable and efficient approach for handling large volumes of issue data across complex repositories. By automating key stages of issue processing, it significantly reduces the need for manual inspection while ensuring consistent and structured analysis. This scalability enables the framework to be effectively applied to both small and large-scale projects without substantial performance degradation.

In addition, the system enhances the bug triaging process by filtering out low-quality and noisy reports, detecting duplicate issues, and prioritizing actionable bug reports based on their relevance and completeness. This structured approach allows project maintainers to quickly identify and focus on critical issues, thereby improving the efficiency and effectiveness of issue resolution workflows.

Furthermore, by reducing redundancy and organizing issue data more systematically, the framework contributes to improved developer productivity. Developers can spend less time on manual triaging and administrative tasks, and more time on debugging, feature development, and system improvement. This leads to more streamlined maintenance processes and better utilization of development resources.

Overall, the proposed framework demonstrates the practical value of repository mining for software quality assessment. The generated datasets and analytical insights can support future research in areas such as defect prediction, automated triaging, and software analytics. As open-source ecosystems continue to expand, such intelligent and scalable tools will play a crucial role in maintaining software reliability, improving collaboration, and enhancing developer efficiency.

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